

4.6.1 Forces and energy changes

By representing forces as vectors it is possible to explain how objects interact by a variety of contact and non-contact forces. Work is introduced as an important means of energy transfer. A force acts on an object with mass when in a gravitational field and thus distinguishes between mass and weight. An object gains potential energy when raised in a gravitational field because of the work done. Forces can stretch, bend or compress objects. The deformation may be elastic or inelastic depending on the material and the size of the forces involved. The work done in stretching can be used to calculate the potential energy of a spring.

The required practical is an investigation of the relationship between force and extension for a spring.

4.6.1.1 Forces as vectors

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall examples of ways in which objects interact: by gravity, electrostatics, magnetism and by contact (including normal contact force and friction), and describe how such examples involve interactions between pairs of objects which produce a force on each object, representing such forces as vectors.	Scalar quantities have magnitude only. Vector quantities have magnitude and an associated direction. Force is a vector quantity. A vector quantity may be represented by an arrow. The length of the arrow represents the magnitude, and the direction of the arrow the direction of the vector quantity. A force is a push or pull that acts on an object due to the interaction with another object. All forces between objects are either: • contact forces – the objects are physically touching or • non-contact forces – the objects are physically separated.	

4.6.1.2 Resolving forces (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe examples of the forces acting on an isolated solid object or system; describe, using free body diagrams, examples where several forces lead to a resultant force on an object and the special case of balanced forces when the resultant force is zero (qualitative only).	A number of forces acting on an object may be replaced by a single force that has the same effect as all the original forces acting together. This single force is called the resultant force. A free body diagram shows the magnitude and direction of the forces acting on an object. A single force can be resolved into two components acting at right angles to each other. The two component forces together have the same effect as the single force.	WS 1.2, MS 4a, 5a, 5b Use vector diagrams to illustrate resolution of forces and equilibrium situations and determine the resultant of two forces, to include both magnitude and direction (scale drawings only).

4.6.1.3 Work

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe and calculate the changes in energy involved when a system is changed by the work done by forces acting upon it. Use the relationship between work done, force and distance moved along the line of action of the force, describing the energy transfer involved.	A force does work on an object when the force causes a displacement of the object. work done = force $\times$ distance (moved along the line of action of the force) [$W = F s$] work done, W, in joules, J force, F, in newtons, N distance, s, in metres One joule of work is done when a force of one newton causes a displacement of one metre. 1 joule = 1 newton-metre	WS 1.2, MS 3b, 3c Recall and apply this equation to calculate energy transfers. WS 4.5, MS 1c, 3c Convert between newton-metres and joules.

4.6.1.4 Mass and weight

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Define weight, describe how it is measured and describe the relationship between the weight of that body and the gravitational field strength.	Weight is the force acting on an object due to gravity. The force of gravity close to the Earth is due to the gravitational field around the Earth. The weight of an object depends on the gravitational field strength at the point where the object is: $\text{weight} = \text{mass} \times \text{gravitational field strength}$ $[W = m g]$ weight, W, in newtons, N mass, m, in kilograms, kg gravitational field strength, g, in newtons per kilogram, N/kg The weight of an object and the mass of an object are directly proportional. Weight is measured using a calibrated spring balance (a newtonmeter).	WS 1.2, MS 3b, 3c Recall and apply this equation. In any calculation the value of the gravitational field strength (g) will be given. MS 3a Understand and use the symbol for proportionality, $\propto$

4.6.1.5 Gravitational potential energy

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate the amounts of energy associated with an object raised above ground level.	An object raised above ground level gains gravitational potential energy $g.p.e. = \text{mass} \times \text{gravitational field strength} \times \text{height}$ $E_p = m g h$ gravitational potential energy, E_p, in joules, J mass, m, in kilograms, kg gravitational field strength, g, in newtons per kilogram, N/kg height, h, in metres, m	WS 1.2, MS 3c Recall and apply this equation to calculate changes in stored energy. In any calculation the value of the gravitational field strength (g) will be given.

4.6.1.6 Elastic deformation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain, with examples, that to stretch, bend or compress an object, more than one force has to be applied. Describe the difference between elastic and inelastic distortions caused by stretching forces; describe the relationship between force and extension for a spring and other simple systems; describe the difference between linear and non-linear relationships between force and extension, and calculate a spring constant in linear cases.	An object that has been stretched has been elastically deformed if the object returns to its original length after the forces are removed. An object that does not return to its original length after the forces have been removed has been inelastically deformed. The extension of an elastic object, such as a spring, is directly proportional to the force applied, provided that the limit of proportionality is not exceeded. <i>force = spring constant $\times$ extension</i> $[F = k e]$ force, F, in newtons, N spring constant, k, in newtons per metre, N/m extension, e, in metres, m A force that stretches (or compresses) a spring does work, and elastic potential energy is stored in the spring. Provided the spring does not go past the limit of proportionality the work done on the spring and the elastic potential energy stored are equal.	WS 1.2, MS 3c, 4a, 4b, 4c Recall and apply this equation

Required practical activity 13: investigate the relationship between force and extension for a spring.

AT skills covered by this practical activity: physics AT 1 and 2.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 169)

4.6.1.7 Energy stored in a stretched spring

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate the amounts of energy associated with a stretched spring.	Elastic potential energy is stored in a stretched spring. <i>elastic potential energy</i> = 0.5 × <i>spring constant</i> × (<i>extension</i>)² $[E_e = \frac{1}{2} k e^2]$ (assuming the limit of proportionality has not been exceeded) elastic potential energy, E_e, in joules, J spring constant, k, in newtons per metre, N/m extension, e, in metres, m	WS 1.2, MS 1c, 3c Calculate the work done in stretching. MS 3b, 3c Apply this equation, which is given on the Physics equations sheet.